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Vehicle hitch mountable device for lifting objects

Abstract

The present disclosure relates a device for lifting objects. The device comprises a pair of arms, a horizontal bar, a slide bar, and a housing. The pair of arms are configured to be in an open state to lift an object and in a closed state when the pair of arms are not being used. The device is configured to be elongated to a length for providing ease of loading and unloading of an object. The device further comprises a locking mechanism for providing stability during transit. The housing comprises a roller over a screw for providing easy displacement of the slide bar with respect to the housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application hereby claims priority to and incorporates by reference the entirety of the disclosures of the provisional application No. 63/317,826 entitled “VEHICLE HITCH MOUNTABLE DEVICE FOR LIFTING OBJECTS” filed on 8 Mar. 2022.

FIELD

(1) The present disclosure generally relates to a device for lifting objects. More particularly, the present disclosure relates to a device attachable to a vehicle hitch receiver and adapted for lifting objects.

BACKGROUND

(2) The use of personal vehicles for transportation is quite widespread. Recently, people have started to use their personal vehicles for long-distance and long-duration travel instead of travelling by airplanes or trains. In such situations, the luggage tends to be more. People generally use overhead carrier or storage compartments inside the vehicle for carrying their luggage. However, the use of overhead carrier requires use of securing arrangements which do not necessarily hold the luggage in a secure manner. Further, luggage stored inside a vehicle occupies a lot of space. Hence, these options are not feasible for carrying large luggage. Furthermore, people may want to transport large number of goods, for instance, wholesale goods, as well as heavy objects, on their own instead of hiring transport services to save costs of transportation.

(3) For such reasons, people use attachment of external support structures with the vehicles to carry objects. Such support structure is generally known as hitch. Different types of hitches are available which are attached to hitch receivers at the rear end of vehicles to carry objects. All the hitches have different structures and methods to operate. The hitches generally include carry arms or forks to carry objects. For loading and unloading of objects, the carry arms or forks are displaceable using different mechanisms. Some available hitches use motor-chain assembly or motor-flexible member assembly. The design of such displacement arrangement is very complex with different mechanical parts, which make such hitches difficult to maintain. For example, regular servicing, or oiling, of such mechanical parts is required. Other hitches use inbuilt jacks or motors with battery for such displacement, adding complexity to the design. Further, when not in use, the carry arms or the forks are folded. Different mechanisms for folding the carry arms are used, such as brackets, pins and holes in which arms are detached and holes and brackets are aligned for reattaching the arms in different orientation. In some hitches, hinges are used for folding the arms when they are not used. Such designs for folding the arms are complex and time-consuming.

(4) Furthermore, in case of objects with heavy load, the arms tend to bend due to the load. There is no provision in existing hitches to address such issue. Moreover, in such case of heavy load, more energy is required in conventional hitches to lift the arms. Moreover, stability and secure locking of objects are other non-addressed issues with the existing hitches. Hence, while such support structures or hitches are available, there are limitations associated with the available structures, such as limited object carrying capacity, complex structure, insecure connection, and the like.

SUMMARY

(5) While the way that the present disclosure addresses the disadvantages of the prior art will be discussed in greater detail below, in general, the present disclosure provides a device attachable to a vehicle and configured for lifting objects.

(6) An object of the present disclosure is to provide a device which is simple in structure.

(7) Another object of the present disclosure is to provide a device which is easy to use.

(8) Yet another object of the present disclosure is to provide a device which uses less energy while lifting objects.

(9) Yet another object of the present disclosure is to provide a device which provides greater stability to the objects in transit.

(10) Yet another object of the present disclosure is to provide a device which restricts movement of the objects, thereby providing security of the objects while in transit.

(11) Further object of the present disclosure is to provide a device which provides locking of the device in a specific state to prevent rattles during transit.

(12) Further object of the present disclosure is to provide a device which provides latching of the device in case of carrying objects with heavy load.

(13) Further object of the present disclosure is to provide a device which can be attachable to different types of vehicles.

(14) Further object of the present disclosure is to provide a device which can lift different types of objects.

(15) Further object of the present disclosure is to provide a device which does not require any special skill for operating the device.

(16) Further object of the present disclosure is to provide a device which changes states quickly, and hence, saves time.

(17) Further object of the present disclosure is to provide a device that utilizes electricity, thereby preventing manual labor and/or effort.

(18) The device in accordance with the present disclosure comprises a pair of arms, a horizontal bar, a slide bar, and a housing. The pair of arms are configured to be in an open state while carrying an object and in a closed state while not being used. The pair of arms are coupled with the horizontal bar using a coupling mechanism. The coupling mechanism is present at proximal and

distal portions of the horizontal bar.

(19) In an embodiment, the coupling mechanism comprises a spring plunger pin, a clevis pin, and a cotter pin. The clevis pin and the cotter pin engage with walls of the proximal portion and distal portion of the horizontal bar and the pair of arms to couple the arms with the horizontal bar such that free movement of the pair of arms with respect to the horizontal bar is achieved. Hence, the pair of arms may change state from an open state to a closed state and vice versa. For restricting the pair of arms in the open state or the closed state, the pair of arms comprises a first notch and a second notch for the spring plunger pin to attach therewith. In the open state, the spring plunger pin passes through a hole from the corresponding wall of the horizontal bar and is attached to the first notch. Hence, the movement of the pair of arms is restricted and the pair of arms remains in the open state. To change the state of the pair of arms, the spring plunger pin is pulled and is attached to the second notch on the pair of arms. Hence, the movement of the pair of arms is restricted and the pair of arms remain in the closed state.

(20) In an embodiment, a slide bar is coupled with the horizontal bar at substantially middle. The slide bar is configured to be housed in a housing. The slide bar is configured to be displaced from a retracted state to an elongated state and vice versa. For the same, a threaded rod is attached to the slide bar. By rotating the threaded rod, the slide bar may be displaced to the retracted state and to the elongated state.

(21) In an embodiment, the housing comprises a plurality of walls and a screw and a roller attachment at a proximal end of the housing. The roller is configured to reduce (lessen) a friction applied for the displacement of the slide bar. Hence, less energy is used in displacing the slide bar in the retracted state when an object is carried by the pair of arms. The housing further comprises an attachment mechanism for the threaded rod. The attachment mechanism may comprise a plurality of nuts and bolts, one or more bearing, a top plate, and a hex nut. The hex nut is configured to be placed at a distal end of the housing. The rotation of the threaded rod is achieved by rotating the hex nut of the housing. The hex nut may be rotated manually by a wrench, automatically by a motor, or semi-automatically by a handheld drill motor.

(22) In an embodiment, the horizontal bar comprises wedges at a bottom wall. The wedges are configured to restrict displacement of the slide bar in the housing after a predefined level. The wedges work as an obstacle in the displacement. While transiting in the retracted state, the slide bar reaches till the wedges. The further displacement of the slide bar results in a pushing force on the wedges, resulting in bending the wedges. This action results in secure connection of the slide bar with respect to housing, thereby providing stability while transiting with objects.

(23) In an embodiment, the housing comprises a locking mechanism at a distal portion. When the slide bar is displaced to the retracted state, the locking mechanism may be used to lock the slide bar with the housing. This prevents accidental pulling of the slide bar when an object is placed on the pair of arms.

(24) In an embodiment, the device may be used to load and unload an object when the slide bar is in the elongated state. While transiting, the slide bar may be displaced to the retracted state for stability and security of the object.

(25) In an embodiment, the device comprises an insert for attachment of the device with a hitch receiver of vehicles. The insert may be of a rectangular cross-section and may be attached to the hitch receiver by conventional methods.

(26) The present disclosure overcomes the drawbacks of the existing devices by providing a simple and easy to use device for lifting and transiting objects from one place to another. Further, the device provides greater stability to the objects. The device further ensures safe transit by preventing rattles during transit. Furthermore, because of use of the threaded rod and the roller, energy required for the displacement of the pair of arms is very less.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete understanding of the present disclosure may be derived by referring to the detailed description and claims when considered in connection with the Figures, wherein like reference numerals refer to similar elements throughout the Figures, and
- (2) FIG. 1 illustrates a perspective view of a device for lifting objects according to one embodiment of the present disclosure;
- (3) FIG. 2 illustrates an exemplary coupling mechanism when a pair of arms are in an open state, in accordance with an embodiment of the present disclosure;
- (4) FIG. 3 illustrates an exemplary coupling mechanism when a pair of arms are in a closed state, in accordance with an embodiment of the present disclosure;
- (5) FIG. 4 illustrates an exemplary rear perspective view of the coupling mechanism of FIG. 2 and FIG. 3;
- (6) FIG. 5 illustrates an exemplary side view of a device having the pair of arms in the closed state, in accordance with an embodiment of the present disclosure;
- (7) FIG. 6 illustrates an exemplary exploded view of the coupling mechanism in accordance with an embodiment of the present disclosure;
- (8) FIG. 7 illustrates an exemplary rear perspective view of a horizontal bar of a device in accordance with an embodiment of the present disclosure;
- (9) FIG. 8 illustrates an exemplary slide bar in accordance with an embodiment of the present disclosure;
- (10) FIG. 9 illustrates an exemplary wedge in accordance with an embodiment of the present disclosure;
- (11) FIG. 10 illustrates an exemplary slide bar in a retracted state in accordance with an embodiment of the present disclosure;
- (12) FIG. 11 illustrates an exemplary slide bar in an elongated state in accordance with an embodiment of the present disclosure;
- (13) FIG. 12 illustrates an exemplary threaded rod and an internal structure of a housing in accordance with an embodiment of the present disclosure;
- (14) FIG. 13 illustrates an exemplary exploded view of a threaded rod arrangement for the threaded rod of FIG. 12;
- (15) FIGS. 14-FIG. 18 illustrate an exemplary locking mechanism in accordance with an embodiment of the present disclosure; and
- (16) FIG. 19 illustrates an exemplary exploded view of the locking mechanism of FIG. 14-FIG. 18.
- (17) FIG. 20 illustrates a perspective view of a device for lifting objects according to yet another embodiment of the present disclosure;
- (18) FIG. 21 illustrates an exemplary threaded rod and an internal structure of a housing in accordance with another embodiment of the present disclosure;
- (19) FIG. 22 illustrates an exemplary exploded view of a threaded rod arrangement for the threaded rod of FIG. 21;
- (20) FIG. 23 illustrates a perspective view of the device of FIG. 20 connected to a motor vehicle; wherein the slide bar is positioned in an elongated state; and
- (21) FIG. 24 illustrates a perspective view of the device of FIG. 20 connected to a motor vehicle; wherein the slide bar is positioned in a retracted state, and the device is carrying a motorcycle.

DETAILED DESCRIPTION

- (22) The following description is of exemplary embodiments of the invention only, and is not intended to limit the scope, applicability, or configuration of the invention. Rather, the following description is intended to provide a convenient illustration for implementing various embodiments

of the invention. As will become apparent, various changes may be made in the function and arrangement of the elements described in these embodiments without departing from the scope of the invention as set forth herein. It should be appreciated that the description herein may be adapted to be employed with alternatively configured devices having different shapes, components, attachment mechanisms and the like and still fall within the scope of the present invention. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation.

(23) Reference in the specification to “one embodiment” or “an embodiment” is intended to indicate that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least an embodiment of the invention. The appearances of the phrase “in one embodiment” or “an embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

(24) Reference is initially made to FIG. **1** which illustrates a perspective view of the device **10**, in accordance with an embodiment of the present disclosure. The device **10** comprises a pair of arms **100** to provide support to objects and for lifting the objects. The term ‘objects’ as used in the present disclosure is intended to items that are to be lifted and carried, such as heavy luggage, a toolbox, a bike, a pallet, a wheelchair, a camping gear, a hunting gear and the like.

(25) In the illustrated embodiment, each arm of the pair of arms **100** is a four walled structure having a rectangular cross-section. That is, each arm of the pair of arms **100** comprises a first wall **101**, a second wall **102**, a third wall **103** opposite to the first wall **101**, and a fourth wall **104** opposite to the second wall **102**. The first wall **101**, the second wall **102**, the third wall **103**, and the fourth wall **104** are attached with each other so as to form the arms having a rectangular cross section. In some embodiments, a width of each wall forming the arms is about 2 inches, thereby the arms assuming a squared configuration. In some embodiments, a length of each of the pair of arms **100** is about 21 inches. As used herein, the term ‘about’ is meant to encompass deviation of +10% from the specifically mentioned value of a parameter, such as length and width.

(26) The device **10** comprises a horizontal bar **200** comprising a proximal portion **201** and a distal portion **202**. Each of the pair of arms **100** is configured to be coupled with the horizontal bar **200**, in that, one arm of the pair of arms **100** is configured to be coupled to the proximal portion **201** of the horizontal bar **200** and the other arm of the pair of arms **100** is configured to be coupled to the distal portion **202** of the horizontal bar **200**. Each of the pair of arms **100** comprises a distal end **105** at which the arms are coupled to the horizontal bar **200**. In the illustrated embodiment, for coupling of the arms and the horizontal bar **200**, each of the pair of arms **100** comprises a hollow passage, as will be described in detail further below.

(27) Each of the pair of arms **100** are configured to shift between an open state and a closed state. The open state of the pair of arms **100** is a state in which the pair of arms **100** are placed substantially parallel to the ground and are configured to carry an object thereon, for instance, as illustrated in FIG. **1**. In other words, the device **10** is configured to support an object placed on the pair of arms **100**, and further, lift the object placed on the pair of arms **100** when the pair of arms **100** are in the open state. In the closed state, the pair of arms **100** are substantially perpendicular to the ground. In other words, when the device **10** is not being used to lift an object, the pair of arms **100** are configured to be in the closed state. The transition of each of the pair of arms **100** between the open state and the closed state is achieved by a coupling mechanism **207**. The device **10** comprises a coupling mechanism at both the proximal portion **201** and the distal portion **202** of the horizontal bar **200** to attach the pair of arms **100**. In some embodiments, each of the pair of arms **100** is configured to be in an individual open state and an individual closed state and is configured to shift between the states using the corresponding coupling mechanism.

(28) FIGS. **2-6** illustrate an exemplary coupling mechanism **207** in accordance with an embodiment of the present disclosure. FIG. **2** illustrates a coupling mechanism **207** at the proximal portion **201** of the horizontal bar **200** having the corresponding arm of the pair of arms **100** in the open state. FIG. **3** illustrates the coupling mechanism **207** at the proximal portion **201** of the horizontal bar **200**

having the corresponding arm of the pair of arms **100** in the closed state.

(29) The coupling mechanism **207** is configured to couple the pair of arms **100** with the horizontal bar **200** and facilitate transition of the pair of arms **100** from the open state to the closed state, and vice versa. The coupling mechanism **207** comprises a spring plunger pin **271**, a clevis pin **272**, and a cotter pin **273** (seen in FIG. **4**). In order for more effective functionality of the coupling mechanism **207**, a coupling structure **208** may be provided at the proximal portion **201** and the distal portion **202** of the horizontal bar **200**. As seen in FIGS. **2-3**, the coupling structure **208** comprises an outer surface **281** and an inner surface **290** formed at the proximal portion **201** of the horizontal bar **200**. It is appreciated that the outer surface **281** and the inner surface **290** are spaced from each other to allow a corresponding arm to be received therein. That is, dimensions of coupling structure **208** at the proximal portion **201** corresponds to dimension of a corresponding arm of the pair of arms **100**, so as to receive each of the corresponding arm therewithin.

(30) In order to facilitate coupling of the corresponding arm to the horizontal bar **200**, each of the outer surface **281** and the inner surface **290** of the coupling structure **208** comprises a hole **285** (best seen in FIG. **6**) corresponding to the hollow passage **111** provided at the corresponding arm of the pair of arms **100**. In other words, the first hollow passage **111** provided at the distal end **105** of the corresponding arm of the pair of arms **100** coincides with the hole **285** of the inner surface **290** and outer surface **281** of the coupling structure **208** at the proximal portion **201** of the horizontal bar **200**. By virtue of the hollow passage coinciding with holes in the surfaces of the coupling structure **208**, a single pin can be passed therethrough for coupling of the corresponding arm to the proximal portion **201** of the horizontal bar **200**.

(31) For instance, for coupling of the corresponding arm with the proximal portion **201** of the horizontal bar **200**, the clevis pin **272** is passed through the respective holes in the corresponding arm and the coupling structure, and further, the cotter pin **273** is engaged at a through hole of the clevis pin **272** for ensuring the coupling of the corresponding arm with the horizontal bar **200**. FIG. **4** illustrates an exemplary rear perspective view of the corresponding arm coupled to the proximal portion **201** of the horizontal bar **200** with the clevis pin **272** and the cotter pin **273**. The use of clevis pin **272** and the cotter pin **273** facilitates easy and simple coupling between the arm and the horizontal bar **200**. Further, a flexible movement of the arm is achieved through such coupling.

(32) It is to be noted that although the coupling mechanism **207** and coupling structure **208** has been described with reference to the proximal portion **201**, the distal portion **202** of the horizontal bar **200** comprises a similar coupling mechanism **207** and coupling structure **208** in order to couple the other arm of the pair of arms **100** with the horizontal bar **200**.

(33) For locking the pair of arms **100** in the open state or the closed state, the outer surface of the coupling structure **208** at proximal portion **201** of the horizontal bar **200** comprises an opening **284** (best seen in FIG. **6**) to pass the spring plunger pin **271** therethrough. For locking the pair of arms **100** in the open state, the corresponding arm comprises a first notch **113** (best seen in FIG. **3**) for receiving the spring plunger pin **271** in the open state. The first notch **113** can be provided at the first wall **101** of the corresponding arm. Similarly, for locking the corresponding arm in the closed state, the first wall **101** of the corresponding arm comprises a second notch **114** (best seen in FIG. **2**) for receiving the spring plunger pin **271** in the closed state. The first notch **113** and the second notch **114**, on the first wall of the corresponding arms, can be placed at a suitable position with respect to the hole of the clevis pin **272**. For changing a state of the corresponding arm, the spring plunger pin **271** is pulled and inserted into a desired notch. For example, for changing the state of the corresponding arm from the open state to the closed state, the spring plunger pin **271** is pulled and inserted into the second notch **114**. Similarly, for changing the state of the corresponding arm from the closed state to the open state, the spring plunger pin **271** is pulled and inserted into the first notch **113**. Hence, the first notch **113** is configured to restrict movement of the arm in the open state, and the second notch **114** is configured to restrict movement of the arm in the closed state. Such restriction of movement in the open state prevents accidental closure of the pair of arms **100**

when in the open state and facilitates stability to the device **10** in the open state as well as the closed state. It is to be noted that the device **10** comprises a similar coupling mechanism **207** at the distal portion **202** of the horizontal bar **200**. Hence, the coupling mechanism **207** is configured to control movement of the pair of arms **100**. Accordingly, the coupling mechanism **207** provides advantage of quick folding and unfolding of the pair of arms **100**. Hence, a process of changing states of the pair of arms **100** is quick and easy, and hence, saves time.

(34) FIG. 5 illustrates side view of the device **10** having the pair of arms **100** in the closed state. As shown in FIG. 5, the pair of arms **100**, in the closed state, makes an angle α with respect to a line parallel to the ground. In some embodiments, the pair of arms **100** are perpendicular to the ground, and hence, the angle α is 90 degrees. In other embodiments, the pair of arms **100** are substantially perpendicular to the ground making an obtuse angle with respect to the ground. In such embodiments, the angle α is more than 90 degrees, and more particularly, 93 degrees.

(35) Referring to FIG. 7, an exemplary horizontal bar **200** is illustrated. The horizontal bar **200** comprises a top wall **203**, a bottom wall **204**, and a front wall **205**. In some embodiments, the top wall **203**, the bottom wall **204**, and the front wall **205** are substantially straight and are configured to extend between the proximal portion **201** to the distal portion **202** of the horizontal bar **200**. In some embodiments, a back portion of the horizontal bar **200** is uncovered. In other embodiments, a plurality of supports **206** may be disposed between the top wall **203** and the bottom wall **204** of the horizontal bar **200** to provide stability.

(36) The device **10** further comprises a slide bar **300** attached to the horizontal bar **200**. An exemplary slide bar **300** in accordance with the present disclosure is illustrated in FIG. 8. The slide bar **300** is configured to be in an elongated state and a retracted state. The slide bar **300** comprises a front surface **301**, a back surface **302**, and a plurality of side surfaces **303** attached in a way so as to form a rectangular cross-section. In an embodiment, the slide bar **300** is a hollow structure. The slide bar **300** is configured to be attached to the horizontal bar **200**. In a preferred embodiment, the slide bar **300** is attached to the horizontal bar **200** at a substantially middle portion of the horizontal bar **200**. Such attachment facilitates substantially equal distribution of weight of the objects at both sides of the horizontal bar **200**. In an embodiment, the slide bar **300** is attached to the horizontal bar **200** by means of welding. Any other method for attaching the slide bar **300** with the horizontal bar **200** known in the art may be used.

(37) Referring back to FIG. 1, the device **10** further comprises a housing **400** configured to enclose the slide bar **300** therewithin. In an embodiment as seen in FIG. 9, the housing **400** is hollow, and comprises a front wall **401**, a back wall **402**, and a plurality of side walls **403**. A dimension of the housing **400** corresponds to that of the slide bar **300**. The housing **400** is configured to displace the slide bar **300** in the vertical direction. Accordingly, the slide bar **300** is configured to change states with respect to the housing **400**.

(38) The slide bar **300** is configured to be in the retracted state when it is at least substantially housed within the housing **400** and in the elongated state when it is at least substantially displaced out of the housing **400**. With the displacement of the slide bar **300**, the horizontal bar **200**, and hence, the pair of arms **100** displace along with the slide bar **300**. In the elongated state of the slide bar **300**, the pair of arms **100** are configured to be closest to the ground. Hence, in a preferred embodiment, the elongated state of the slide bar **300** is used for loading an object on the pair of arms **100**. Similarly, the elongated state is further used for unloading an object. In an exemplary embodiment, a total height of the device **10** when the slide bar **300** is in the retracted state is around 26 inches. FIG. 10 illustrates the slide bar **300** at least substantially housed in the housing **400**, and hence, in the retracted state. FIG. 11 illustrates the slide bar **300** in the elongated state. In the elongated state, the total height of the device **10** when the slide bar **300** is around 42 inches. Hence, from the retracted state to the elongated state, the height of the device **10** is increased by approximately 16 inches. Thus, the device **10** may be used to lift an object having varying height till 42 inches. In an embodiment, an outer diameter of the device **10** is 33 inches.

(39) Referring back to FIG. 7, the housing **400** comprises a smoothening structure at a proximal end thereof to provide smooth and minimal friction (friction less) displacement of the slide bar **300** with respect to the housing **400**. For instance, the smoothening structure may comprise a shoulder screw **405** and a roller **406**. The shoulder screw **405** is passed through a through provision **404** (best seen in FIG. 9) on the plurality of side walls **403** of the housing **400**. The roller **406** is configured to be placed over the shoulder screw **405**. The roller **406** is placed such that during displacement of the slide bar **300**, the roller **406** comes in contact with the back surface **302** of the slide bar **300**. In the elongated state, when an object is placed over the pair of arms **100**, a large amount of weighing force is levied by the weight of the object at a point where the slide bar **300** is attached to the horizontal bar **200**. At this stage, an energy required for the displacement of the slide bar **300** into the retracted state is very high. However, due to the placement of the roller **406**, a rolling force is applied while displacing, which reduces (lessens) the friction at the time of displacement, and hence, the energy required for such displacement is significantly less as compared to the energy required in absence of the roller **406**. This results in smooth displacement of the slide bar **300** even with an object having significant weight.

(40) Referring to FIG. 9, the device **10** further comprises a wedge **209** on an inner surface of the bottom wall **204** of the horizontal bar **200** adjacent to each of the side surfaces **303** of the slide bar **300**. The wedge **209** is of a predefined height. In an embodiment, in the retracted state of the slide bar **300**, the roller **406** is configured to be placed adjacent to the wedge **209**. While displacing the slide bar **300** into the retracted state, at a predefined level L, a lower edge **407** of the housing **400** coincides with an upper edge **292** of the wedge **209** at each of the side walls **403** of the housing **400**, which restricts further movement of the slide bar **300** from the predefined level L. At this stage when the lower edge **407** of the housing **400** coincides with the upper edge **292** of the wedge **209**, the retracted state is achieved. Hence, the wedge **209** works as an obstacle for displacement of the slide bar **300** beyond the predefined level L and the slide bar **300** is not able to displace further. The further displacement of the slide bar **300** results in a pushing force on the wedge **209**. This action bends the wedge **209** to a certain angle such that the wedge **209** makes an acute angle with the bottom wall **204** of the horizontal bar **200** and with the ground, resulting in a very large binding force between the slide bar **300** and the horizontal bar **200**. Hence, the wedge **209** provides stability to the device **10**. FIG. 9 illustrates the pushing forward action indicated by an arrow A. The bended wedge **209** provides secure locking of the slide bar **300** in the retracted state to prevent rattles during transit of the object using the device **10**.

(41) Referring to FIG. 12, and also shown in FIG. 11, the device **10** comprises a threaded rod **500** configured to pass within the slide bar **300** and the housing **400**. The threaded rod **500** is configured to be rotated in the clockwise direction and the anticlockwise direction. The rotation of the threaded rod **500** results in displacement of the slide bar **300** with respect to the housing **400**. In an embodiment, threads of the threaded rod **500** are acme threads. The acme threads facilitate the threaded rod **500** to rotate smoothly and with less energy, which makes the displacement of the slide bar **300** smooth and easy. In the elongated state, the threaded rod **500** is configured to be housed within the housing **400** and partially in the slide bar **300**. In the retracted state, the threaded rod **500** is configured to be housed within the housing **400** and at least substantially within the slide bar **300**. The threaded rod **500** is attached with the slide bar **300** by means of a coupling plate **304** and a nut **305** coupled to an upper surface of the coupling plate **304**. The coupling plate **304** and the nut **305** may be attached by any suitable method known in the art. In an embodiment, the threaded rod **500** is passed through the nut **305** and the coupling plate **304** at the time of assembling the device **10**.

(42) For providing the displacement of the slide bar **300**, the housing **400** comprises a threaded rod arrangement **408**. The threaded rod arrangement **408** is configured to couple the threaded rod **500** with a distal end of the housing **400**. For the threaded rod **500** to be housed within the housing **400**, for instance, the threaded rod arrangement **408** may comprise a threaded shaft collar **481**, a thrust

plain bearing **482**, a top lifting plate **483**, a thrust ball bearing **484**, and a hex nut **485**. FIG. **12** illustrates an exemplary threaded rod arrangement **408** and an internal structure of the distal end of the housing **400** in association with the slide bar **300** while FIG. **13** illustrates an exploded view of the threaded rod arrangement **408**. The hex nut **485** is configured to be placed at a distal end of the threaded rod **500**. The thrust ball bearing **484**, and the thrust plain bearing **482** are configured to provide smooth rotation of the threaded rod **500**, and hence, smooth displacement to the slide bar **300**. For the attachment of the threaded rod **500** with the housing **400**, the top lifting plate **483** is provided, which is attached to a distal end of the housing **400** using a plurality of screws **486** and nuts **487**.

(43) FIG. **14-18** illustrate an exemplary locking mechanism **409** of the housing **400** in accordance with the present disclosure. The locking mechanism **409** is configured to lock the slide bar **300** with the housing **400** in the retracted state to provide stability and to lessen effects of any rattle while carrying an object in the open state, and when the device **10** is not being used, in the closed state. The locking mechanism **409** comprises a latch bolt **491**, a spring **492**, a latch link rod **493**, a latch ferrule **494**, and a ball knob **495**. The locking mechanism **409** further comprises a lock latch **496** and a cam lock **497** (seen in FIGS. **18-19**). The latch bolt **491** comprises a groove at a distal end for accommodating a protrusion of the lock latch **496** therewithin.

(44) FIG. **14** illustrates an exemplary state when the slide bar **300** is displaced to the retracted state. To lock the slide bar **300**, the ball knob **495** is pulled as indicated by an outward arrow, thereby retracting the spring **492**, and the threaded rod **500** is rotated. The rotation of the threaded rod **500** displaces the slide bar **300** towards the distal end of the housing **400** as indicated by an upward arrow. When the slide bar **300** reaches at the predefined distance from the distal end of the housing **400**, the ball knob **495** is released, thereby releasing the spring **492**. Hence, the ball knob **495** reaches at a desired position due to the release of the spring **492** as indicated by an inward arrow for locking the slide bar **300**, as illustrated in FIG. **15**. The lock latch **496** is then rotated in a first direction, indicated by a rotating arrow, by rotating the cam lock **497** with a key. This rotation of the cam lock **497** results in rotation of the protrusion of the lock latch **496** and in engagement of the lock latch **496** to the groove of the latch bolt **491**. This configuration, as illustrated in FIG. **16**, locks the slide bar **300** in the retracted state. When the device **10** is carrying an object and the slide bar **300** is in the retracted state, the locking mechanism **409** holds the slide bar **300** in the retracted state, thereby preventing accidental pulling out of the slide bar **300** in transit. For unlocking and displacement of the slide bar **300**, the lock latch **496** is rotated in a second direction as indicated by a second rotating arrow in FIG. **17**, opposite to the first direction, and the ball knob **495** is pulled in an outward direction, indicated by an outward arrow. This configuration releases the slide bar **300** and facilitates displacement of the slide bar **300** by rotating the threaded rod **500**. FIG. **18** illustrates a top perspective view of the locking mechanism **409** wherein the slide bar **300** is locked with the housing **400**. The locking mechanism **409** thus locks the slide bar **300**. FIG. **19** illustrates an exploded view of the locking mechanism **409** illustrating the various components in accordance with the present disclosure.

(45) The rotation of the threaded rod **500** is controlled by rotating the hex nut **485** at the distal end of the housing **400**. The rotation of the threaded rod **500** may be performed manually, automatically, or semi-automatically. In embodiment as seen in FIGS. **1-19**, the hex nut **485** may be removably attached to an electric actuator such that the threaded rod **500** is configured to be rotated using an electric actuator. The electric actuator can include but not limited to: handheld drill motor (not shown in figures). Thus, the rotation of the threaded rod **500** may be performed using the electric actuator in semi-automatic fashion.

(46) In some embodiments, the threaded rod **500** is rotated manually by attaching a wrench to the hex nut **485**. Accordingly, the operation of the device **10** is very simple. No internal motor, energy source for the battery, or no jack is needed for the displacement. Thus, the design of the device **10** is very simple. Thus, the rotation of the threaded rod **500** may be performed manually.

(47) In yet other embodiments, the rotation of the threaded rod **500** is controlled by attaching a handheld drill motor (not shown in figures) to the hex nut **485**. Hence, the displacement of the slide bar **300** from the retracted state to the elongated state and vice versa, is achieved by simply actuating the handheld drill motor (not shown in figures) attached to the hex nut **485**. Further, the operation of the device **10** is also very easy, which makes the device **10** easy to use by any person as no extra skill is needed to use the device **10**.

(48) Referring back to FIG. **1**, the device **10** further comprises a plurality of strap points to attach a securing mechanism for securing objects with the device **10**. For instance, an eye bolt **600** may be attached to each of the strap points to facilitate attachment of an object with the device **10** using a plurality of strap bands to secure tie and to restrict movement of the object while transiting the object from one place to another. In an exemplary embodiment, the strap points may be provided on the top wall **203** of the horizontal bar **200**. The strap points may be provided at other suitable location as per requirement.

(49) In some embodiments, the device **10** further comprises a second locking mechanism at a preferred location on the housing **400** for displacement of the slide bar **300** to a certain height. In such embodiment, the slide bar **300** may be in a semi-elongated state. This configuration may be used to adjust the total height of the device **10** with respect to a certain height of the object. The height adjustment may result in better lifting and support of the object while in transit.

(50) The device **10** is configured to carry objects having different weights. The device **10**, for instance, is configured to carry an object having weight up to 1000 lbs. Some non-limiting examples of the objects carried by the device **10** are heavy luggage, a toolbox, a bike, a pallet, a wheelchair, a camping gear, a hunting gear and the like.

(51) For attachment of the device **10** with a hitch receiver of the vehicle, the device **10** comprises an insert **700** (seen in FIG. **1**) configured to be inserted into a standard hitch receiver. Generally, the insert **700** is of a rectangular cross-section, and is configured to be inserted and engaged with the hitch receiver at the rear of a vehicle. The insert **700** comprises a pair of walls **701** configured to be placed opposite to each other. In some embodiments, the insert **700** is configured to be placed at the distal end of the housing **400** and adjacent to the screw and the roller **406**, as seen in FIG. **9**. The insert **700** may be attached to the housing **400** by way of welding. Any other conventional methods for attaching the insert **700** with the housing **400** may also be used.

(52) The insert **700** is configured to be engaged to the hitch receiver of all types of vehicles, such as an SUV, a hatchback car, a minibus, a truck, and the like. To be engaged with the hitch receiver, a first pair of walls **701** comprises engagement means **702**. The insert **700** is inserted into the hitch receiver such that the engagement means of the insert **700** is aligned with corresponding engagement means in the hitch receiver. In some embodiments, the engagement means **702** of the insert **700** is an aperture and the engagement means of the vehicle is a bolt, the aperture configured to receive the bolt to engage the insert, and thus, the device **10**, to the hitch receiver of the vehicle.

(53) In some embodiments, at least some of the components of the device **10** are formed using steel. In some embodiments, all of the components of the device **10** are formed using steel.

(54) FIG. **20** illustrates a perspective view of a device **20** for lifting objects according to yet another embodiment of the present disclosure. The device **20** is similar to device **10** (FIGS. **1-19**) except modifications associated with the housing as well as addition of an intermediate arm **100** between pair of arms **100**. The device **20** comprises three arms **100**, that is, one intermediate arm **100** in addition to pair of arms **100** of FIG. **1**, to provide support to objects and for lifting the objects. Further, the device **20** further comprises a support structure **110** connected to each arm **100**. As shown in FIG. **20**, the support structure **110** is substantially ladder-shaped to support weight (load) of objects. The term 'objects' as used in the present disclosure is intended to items that are to be lifted and carried, such as heavy luggage, a toolbox, a bike, a pallet, a wheelchair, a camping gear, a hunting gear, and the like.

(55) The device **20** comprises a horizontal bar **200**, similar to a horizontal bar **200** of FIG. **1**,

comprising a proximal portion **201** and a distal portion **202**. Each of the three arms **100** is configured to be coupled with the horizontal bar **200**, in that, one arm of the three arms **100** is configured to be coupled to the proximal portion **201** of the horizontal bar **200**, intermediate arm of the three arms **100** is configured to be coupled to the intermediate portion **210** of the horizontal bar **200**, and the remaining arm of the three arms **100** is configured to be coupled to the distal portion **202** of the horizontal bar **200**. Each of the three arms **100** comprises a distal end **105** at which the arms are coupled to the horizontal bar **200**. In the illustrated embodiment as shown in FIG. **20**, for coupling of the arms and the horizontal bar **200**, two arms of the three arms **100** coupled to extremities of the horizontal bar **200**, that is the proximal portion **201** and the distal portion **202**, comprises a hollow passage, similar as described earlier in reference to FIG. **6**. In the illustrated embodiment as shown in FIG. **20**, the intermediate arm **100** is fixedly coupled to the intermediate portion **210** of the horizontal bar **200**.

(56) Further, in an exemplary embodiment (not shown in figures), for coupling of the arms and the horizontal bar **200**, each of the three arms **100** comprises a hollow passage, similar as described earlier in reference to FIG. **6**. Each of the three arms **100** are configured to shift between an open state and a closed state. The open state of the three arms **100** is a state in which the three arms **100** are placed substantially parallel to the ground and are configured to carry an object thereon, for instance, as illustrated in FIG. **20**. In other words, the device **20** is configured to support an object placed on the three arms **100**, and further, lift the object placed on the three arms **100** when the three arms **100** are in the open state. In the closed state (not shown in figures), the three arms **100** are substantially perpendicular to the ground. In other words, when the device **20** is not being used to lift an object, the three arms **100** are configured to be in the closed state. The transition of each of the three arms **100** between the open state and the closed state is achieved by a coupling mechanism **207**, similar as described earlier in reference to FIGS. **2-6**. The device **20** comprises a coupling mechanism **207** at proximal portion **201**, intermediate portion **210** and the distal portion **202** of the horizontal bar **200** to attach the three arms **100**. In some embodiments, each of the three arms **100** is configured to be in an individual open state and an individual closed state and is configured to shift between the states using the corresponding coupling mechanism.

(57) In an exemplary embodiment as seen in FIG. **23**, the device **20** further comprises a slide bar **300**, similar to a slide bar **300** of FIG. **8**, attached to the horizontal bar **200**. The slide bar **300** is configured to be in an elongated state and a retracted state.

(58) The device **20** further comprises a housing **400'** configured to enclose the slide bar **300** therewithin. The housing **400'** is similar to housing **400** of FIG. **1**, except modifications associated with threaded rod arrangement **408'**, which will be described in below description. The housing **400'** is configured to displace a slide bar **300**, similar to slide bar **300** of FIG. **8**, in the vertical direction. Accordingly, the slide bar **300** is configured to change states with respect to the housing **400'**.

(59) Referring to FIG. **21**, the device **20** comprises a threaded rod **500** configured to pass within the slide bar **300** and a housing **400'**. The threaded rod **500** is configured to be rotated in the clockwise direction and the anticlockwise direction. The rotation of the threaded rod **500** results in displacement of the slide bar **300** with respect to the housing **400'**. In an embodiment, threads of the threaded rod **500** are acme threads. The acme threads facilitate the threaded rod **500** to rotate smoothly and with less energy, which makes the displacement of the slide bar **300** smooth and easy. In the elongated state, the threaded rod **500** is configured to be housed within the housing **400'** and partially in the slide bar **300** as seen in FIG. **23**. In the retracted state, the threaded rod **500** is configured to be housed within the housing **400'** and at least substantially within the slide bar **300** as seen in FIG. **24**. The threaded rod **500** is attached with the slide bar **300** by means of a coupling plate **304** and a nut **305** coupled to an upper surface of the coupling plate **304**. The coupling plate **304** and the nut **305** may be attached by any suitable method known in the art. In an embodiment, the threaded rod **500** is passed through the nut **305** and the coupling plate **304** at the time of

assembling the device **10**.

(60) For providing the displacement of the slide bar **300**, the housing **400'** comprises a threaded rod arrangement **408'**. The threaded rod arrangement **408'** is similar to threaded rod arrangement **408** (FIGS. **12-13**) except the elimination of hex nut **485** and addition of an electric motor **488** instead of hex nut **485**. The threaded rod arrangement **408'** is configured to couple the threaded rod **500** with a distal end of the housing **400'**. For the threaded rod **500** to be housed within the housing **400'**, for instance, the threaded rod arrangement **408'** may comprise a threaded shaft collar **481**, a thrust plain bearing **482**, a top lifting plate **483**, a thrust ball bearing **484**, and an electric motor **488**. (61) FIG. **21** illustrates an exemplary threaded rod arrangement **408'** and an internal structure of the distal end of the housing **400'** in association with the slide bar **300** while FIG. **22** illustrates an exploded view of the threaded rod arrangement **408'**. The electric motor **488** is configured to be arranged at a distal end of the threaded rod **500** such that a drive shaft (not shown in figures) of the electric motor **488** is rotationally coupled to the distal end of the threaded rod **500**. The thrust ball bearing **484**, and the thrust plain bearing **482** are configured to provide smooth rotation of the threaded rod **500**, and hence, smooth displacement to the slide bar **300**. For the attachment of the threaded rod **500** with the housing **400'**, the top lifting plate **483** is provided, which is attached to a distal end of the housing **400'** using a plurality of screws **486** and nuts **487**.

(62) The rotation of the threaded rod **500** is controlled by actuating the electric motor **488** at the distal end of the housing **400'**. In an embodiment as seen in FIGS. **20-24**, the threaded rod **500** is configured to be rotated using an electric actuator wherein the electric actuator is an electric motor **488**. In an exemplary embodiment, the electric motor **488** could be integrally connected (built-in) to the housing **400'**. In another exemplary embodiment, the electric motor **488** could be retro-fittingly mounted (later installed) to the housing **400'**. The electric motor **488** further comprises an electric connector **488a** such as but not limited to wires, plugs, sockets and combinations thereof. The electric connector **488a** is configured to electrically connect the electric motor **488** with an electrical power source (not shown in figures). In various embodiments, the electrical power source (not shown in figures) could include but not limited to: vehicle tail lights, vehicle cigarette plug (in-car electric socket), vehicle battery, vehicle power outlet, power cables, power sockets available in various premises such as homes, offices, factory and so on. In an exemplary embodiment, the electrical power source (not shown in figures) is retro-fittingly electrically connected to the electric motor **488** by a vehicle technician. The actuation of the electric motor **488** is controlled by a pair of switches, designated as an uplift switch **488b** and a downlift switch **488c**. When the uplift switch **488b** is manually pressed, then the electric motor **488** is rotated in a specific direction to bring the slide bar **300** in the retracted state as shown in FIG. **24**. When the downlift switch **488c** is manually pressed, then the electric motor **488** is rotated in the opposite direction to bring the slide bar **300** in the elongated state as shown in FIG. **23**. Hence, the displacement of the slide bar **300** from the retracted state to the elongated state and vice versa, is achieved automatically by simply actuating the electric motor **488**. Accordingly, the operation of the device **20** is very simple. Further, the operation of the device **20** is also very easy, which makes the device **20** easy to use by any person as no extra skill is needed to use the device **20**.

(63) FIG. **23** illustrates a perspective view of the device **20** (FIG. **20**) wherein the device **20** is connected to a motor vehicle. For attachment of the device **20** with a hitch receiver of the vehicle, the device **20** comprises an insert **700** (seen in FIG. **20**) configured to be inserted into a standard hitch receiver of a motor vehicle. The slide bar **300**, similar to slide bar **300** of FIG. **8**, is configured to be in the retracted state when it is at least substantially housed within the housing **400'** and in the elongated state when it is at least substantially displaced out of the housing **400'**. With the displacement of the slide bar **300**, the horizontal bar **200**, and hence, the three arms **100** displace along with the slide bar **300**. In the elongated state of the slide bar **300**, the three arms **100** are configured to be closest to the ground. Hence, in a preferred embodiment, the elongated state of the slide bar **300** is used for loading an object on the three arms **100**. Similarly, the elongated state is

further used for unloading an object. In an exemplary embodiment, a total height of the device **20** when the slide bar **300** is in the retracted state is around 26 inches. FIG. **24** illustrates the slide bar **300** at least substantially housed in the housing **400'**, and hence, in the retracted state. Further, as seen in FIG. **24**, the device **20** is carrying a motorcycle.

(64) FIG. **23** illustrates the slide bar **300** in the elongated state. In the elongated state, the total height of the device **20** when the slide bar **300** is around 42 inches. Hence, from the retracted state to the elongated state, the height of the device **20** is increased by approximately 16 inches. Thus, the device **20** may be used to lift an object having varying height till 42 inches. In an embodiment, an outer diameter of the device **20** is 33 inches.

(65) It should be well understood that the device **10** (FIGS. **1-19**) and the device **20** (FIGS. **20-24**) and their associated parts and components are very similar and could be interchangeably used. Thus, various embodiments associated with the device **10** (FIGS. **1-19**) could also be used (applicable) in conjunction with the device **20** (FIGS. **20-24**) and vice-versa.

(66) Finally, while the present invention has been described above with reference to various exemplary embodiments, many changes, combinations, and modifications may be made to the exemplary embodiments without departing from the scope of the present invention. For example, the various components may be implemented in alternative ways. These alternatives can be suitably selected depending upon the particular application or in consideration of any number of factors associated with the operation of the device. In addition, the techniques described herein may be extended or modified for use with other types of devices. These and other changes or modifications are intended to be included within the scope of the present invention.

Claims

1. A device (**10,20**) for lifting one or more objects, the device comprising: at least a pair of arms (**100**) configured to receive one or more objects; a horizontal bar (**200**) coupled with the pair of arms (**100**); a slide bar (**300**) coupled with the horizontal bar (**200**), wherein the slide bar (**300**) is configured to be displaced with respect to a housing (**400,400'**) between a retracted state and an elongated state; a threaded rod (**500**) configured to pass within the slide bar (**300**) and the housing (**400, 400'**); wherein the threaded rod (**500**) is configured to be rotated in the clockwise direction and the anticlockwise direction; and rotation of the threaded rod (**500**) displaces the slide bar (**300**) with respect to the housing (**400,400'**); wherein the housing (**400, 400'**) comprises a threaded rod arrangement (**408, 408'**) configured to couple the threaded rod (**500**) with a distal end of the housing (**400, 400'**); an insert (**700**) coupled with the housing (**400,400'**), wherein the insert (**700**) is configured to be engaged to a hitch receiver of a vehicle; and wherein the threaded rod arrangement (**408, 408'**) comprises a threaded shaft collar (**481**), a thrust plain bearing (**482**), a top lifting plate (**483**), and a thrust ball bearing (**484**).
2. The device (**10,20**) according to claim 1, wherein the object comprises at least one of: a heavy luggage, a toolbox, a bike, a pallet, a wheelchair, a camping gear and a hunting gear.
3. The device (**10,20**) according to claim 1, wherein each of the pair of arms (**100**) are configured to shift between an open state and a closed state; wherein in the closed state, the pair of arms (**100**) are substantially perpendicular to the ground; and wherein in the open state, the pair of arms (**100**) are placed substantially parallel to the ground and the pair of arms (**100**) are configured to carry at least one object.
4. The device (**10,20**) according to claim 3, wherein transition of each of the pair of arms (**100**) between the open state and the closed state is achieved by a coupling mechanism (**207**), wherein coupling mechanism (**207**) is configured to couple the pair of arms (**100**) with the horizontal bar (**200**), and the coupling mechanism (**207**) comprises a spring plunger pin (**271**), a clevis pin (**272**) and a cotter pin (**273**).
5. The device (**10,20**) according to claim 1, wherein the device (**10,20**) further comprises a wedge

(209) on an inner surface of a bottom wall (204) of the horizontal bar (200); wherein the wedge (209) limits the displacement of the slide bar (300) and further provides secure locking of the slide bar (300) in the retracted state to prevent rattles during transit of the object using the device (10, 20).

6. The device (10,20) according to claim 1, wherein the threaded rod arrangement (408) further comprises a hex nut (485), wherein the hex nut (485) is configured to be rotated to control rotation of the threaded rod (500).

7. The device (10,20) according to claim 1, wherein the threaded rod arrangement (408') further comprises an electric motor (488) configured to be actuated to control rotation of the threaded rod (500), wherein the electric motor (488) is rotationally coupled to the distal end of the threaded rod (500).

8. The device (10,20) according to claim 1, wherein the housing (400) comprises a locking mechanism (409) configured to lock the slide bar (300) with the housing (400) when the slide bar (300) is in retracted state, wherein the locking mechanism (409) comprises: a latch bolt (491), a spring (492), a latch link rod (493), a latch ferrule (494), a ball knob (495), a lock latch (496) and a cam lock (497).

9. The device (10,20) according to claim 1, wherein the housing (400,400') further comprises a smoothening structure that comprises a roller (406), wherein during displacement of the slide bar (300) with respect to the housing (400,400'), the roller (406) comes in contact with the slide bar (300); wherein the smoothening structure is configured to enable reduced-friction displacement of the slide bar (300) with respect to the housing (400,400').

10. The device (10,20) according to claim 1, wherein the slide bar (300) is configured to be in the retracted state when it is at least substantially housed within the housing (400); and the slide bar (300) is configured to be in the elongated state when it is at least substantially displaced out of the housing (400).

11. A device (10, 20) for lifting one or more objects, the device comprising: at least a pair of arms (100) configured to receive one or more objects; a horizontal bar (200) coupled with the pair of arms (100); a slide bar (300) coupled with the horizontal bar (200), wherein the slide bar (300) is configured to be displaced with respect to a housing (400,400') between a retracted state and an elongated state; a threaded rod (500) configured to pass within the slide bar (300) and the housing (400,400'); wherein the threaded rod (500) is rotatable in clockwise direction and counterclockwise direction; wherein the threaded rod (500) is configured to be rotated by an electric actuator; wherein the rotation of the threaded rod (500) displaces the slide bar (300) with respect to the housing (400,400'); an insert (700) coupled with the housing (400,400'), wherein the insert (700) is configured to be engaged to the hitch receiver of a vehicle; a wedge (209) on an inner surface of a bottom wall (204) of the horizontal bar (200); and wherein the wedge (209) limits the displacement of the slide bar (300) and further provides secure locking of the slide bar (300) in the retracted state to prevent rattles during transit of the object using the device (10, 20).

12. The device (10,20) according to claim 11, wherein the object comprises at least one of: a heavy luggage, a toolbox, a bike, a pallet, a wheelchair, a camping gear, and a hunting gear.

13. The device (10,20) according to claim 11, wherein each of the pair of arms (100) are configured to shift between an open state and a closed state; wherein in the closed state, the pair of arms (100) are substantially perpendicular to the ground and; wherein in the open state, the pair of arms (100) are placed substantially parallel to the ground and the pair of arms (100) are configured to carry at least one object.

14. The device (10,20) according to claim 13, wherein transition of each of the pair of arms (100) between the open state and the closed state is achieved by a coupling mechanism (207), wherein coupling mechanism (207) is configured to couple the pair of arms (100) with the horizontal bar (200), and the coupling mechanism (207) comprises a spring plunger pin (271), a clevis pin (272) and a cotter pin (273).

15. The device (10,20) according to claim 11, wherein the housing (400, 400') comprises a threaded rod arrangement (408, 408') configured to couple the threaded rod (500) with a distal end of the housing (400, 400').
 16. The device (10,20) according to claim 15, wherein the threaded rod arrangement (408, 408') comprises a threaded shaft collar (481), a thrust plain bearing (482), a top lifting plate (483) and a thrust ball bearing (484).
 17. The device (10,20) according to claim 15, wherein the threaded rod arrangement (408) further comprises a hex nut (485), wherein the hex nut (485) is configured to be rotated by using an electric actuator to control rotation of the threaded rod (500).
 18. The device (10,20) according to claim 11, wherein the electric actuator is an electric motor (488) rotationally coupled to the distal end of the threaded rod (500).
 19. The device (10,20) according to claim 17, wherein the electric actuator is a handheld drill motor.
 20. The device (10,20) according to claim 11, wherein the housing (400) comprises a locking mechanism (409) configured to lock the slide bar (300) with the housing (400) when the slide bar (300) is in retracted state, wherein the locking mechanism (409) comprises a latch bolt (491), a spring (492), a latch link rod (493), a latch ferrule (494), a ball knob (495), a lock latch (496) and a cam lock (497).
 21. The device (10,20) according to claim 11, wherein the slide bar (300) is configured to be in the retracted state when it is at least substantially housed within the housing (400); and the slide bar (300) is configured to be in the elongated state when it is at least substantially displaced out of the housing (400).
 22. The device (10,20) according to claim 11, wherein the housing (400,400') further comprises a smoothening structure that comprises a roller (406), wherein during displacement of the slide bar (300) with respect to the housing (400,400'), the roller (406) comes in contact with the slide bar (300); wherein the smoothening structure is configured to enable reduced-friction displacement of the slide bar (300) with respect to the housing (400,400').
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